DSA03 - Natural Language processing

Continuous Internal Assessment - 1

Haritha S
192324299

(1) Answer:

Regular Expression:

Given: L1: Appointment on 05/03/2025 at 10:30.
L2: Call 9876543210 or +91-9123456789.
L3:
L4:

Regular Expression for Date:

```
\b(0[1-9]|[12][0-9]|3[01])/(0[1-9]|1[0-2])/[0-9]{4}\b
```

Requirements:

* days of months
* total months
* Exactly 4-digit year.

\b(0[1-9]) = 1-9 days

[12][0-9] = 10-29 days

3[01] = 30-31 days.

0[1-9] = 1-9 months

1[0-2] = 10-12 months.

[0-9]{4} : Exactly 4-digit year.

Example:

05/03/2025 - Valid

5/3/2025 - Invalid

Regular Expression for Phone Number:

Requirements:

* +91 = optional prefix +91 with hyphen or space.
* 6-9 = starts with 6 and 9.
* Exactly 10-digits
* Optional leading zero.

(?!<\d)(?:?:+91[-]|?:[0])?[6-9][0-9]{9}(?<\d)

(?!<\d) = NO digit immediately after the number.

(?:?:+91[-]) = Indian country code.

[0] = optional leading zero

[6-9][0-9] = Number starts with 6-9.

{9} = Remaining 9 digits.

(?<\9) = NO digit immediately before the number.

Eg: +91-6412356789.
06412356789
6412356789

Regular Expression for Email address:

Requirements:

* user name
* @ symbol
* domain name
* Dot
* specific name ends with atleast 2 letters.

[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}

[A-Za-z0-9._%+-] = user name

@ = @ symbol

[A-Za-z0-9.-] = domain name

\. = Dot

[A,Za-z]{2} = specific name ends with atleast 2 letters.

Example: stephen@edu.in

False Method :

Example: *5876543210 - invalid.

This number is invalid as it starts with 5. A phone number must start with 6-9.

* 9765432101S

This number is Invalid because the phone number has 11 digit, it must contain exactly 10 digit.

Date substitution

plain text : 05/03/2026

Required Input : 2026-03-05

Find pattern :

(\d{2})/(\d{2})/(\d{4})

$3-$2-$1

Python code :

```
import re
text = "Appointment on 05/03/2026 at 10:30"
result = re.sub(
        r'(\d{2})/(\d{2})/(\d{4})',
        r'\g<3>-\g<2>-\g<1>',
        text
)
Print(result)
```

Output : " Appointment on 2026-03-05 at 10:30".

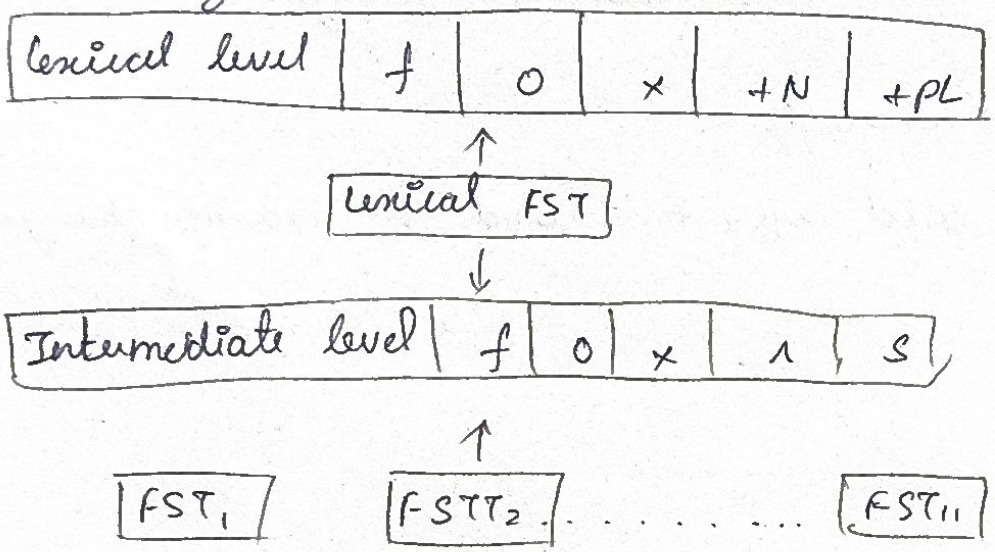
② Finite state morphological Analysis.

Finite state automaten for plural formats:

It has three levels:

* lexical level: root gramatical morpheme.
* Intermediate level: Morphological representations.
* Surface level: Actual word.

Morphological Analysis:



Morpheme Sequence:

Word	Morpheme	Sequence
dogs	dog + s	yes
mice	mouse + s	yes with changes
geese	goose + es	yes with changes
foxes	fox + es	yes

Lexical, Intermediate and Surface form:

Lexical form	Intermediate form	surface form
dog +s	dog +s	dogs
mouse + s	mouse+s	mice

goose + s	goose + es	geese
fox + es	fox + es	foxes

Inflectial and derivational Morphology:

Inflectial: If we add prefix or a suffix to a word the meaning or grammatical function doesn't change.

Derivational Morphology:

If we add suffix to a word the meaning changes.

(3) ii) A training corpus has three sentence:

<s> the cat sat </s> = S1

<s> the cat ran </s> = ~~S3~~

<s> the dog sat </s> = S2

S3 = the dog ran.

Unsmoothed bigram:

$$P(w_i|w_{i-1}) = \frac{c(w_{i-1}, w_i)}{c(w_{i-1})}$$

S1 = "the cat sat"

$P(\text{the}|\langle s\rangle) = \frac{3}{3} = 1$

$P(\text{cat}|\text{the}) = \frac{2}{3}$

$P(\text{sat}|\text{cat}) = \frac{1}{2}$

$P(\langle /s\rangle|\text{sat}) = \frac{2}{2} = 1$

$P(S1) = 1 \times \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{2}{6} = \frac{1}{3}$

$P(S1) = \frac{1}{3} = 0.333$

S2 = "the dog sat"

$P(\text{the}|\langle s\rangle) = \frac{3}{3} = 1$

$P(\text{dog}|\text{the}) = \frac{1}{3}$

$P(\text{sat}|\text{dog}) = \frac{1}{2}$

$P(\langle /s\rangle|\text{sat}) = \frac{2}{2} = 1$

$P(S2) = 1 \times \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{6}$

$P(S2) = 0.166$

S3 = "the dog ran"

$P(\text{the} \mid \overset{<s>}{\text{ran}}) = \frac{3}{3} = 1$

$P(\text{dog} \mid \text{the}) = \frac{2}{3} =$

$P(\text{ran} \mid \text{dog}) = \frac{1}{2}$

$P(<s> \mid \text{ran}) = \frac{1}{2}$

$P(S3) = 1 \times \frac{2}{3} \times \frac{1}{2} \times \frac{1}{2} = \frac{2}{12} = \frac{1}{6}$

$P(S3) = 0.166$

i) Given word: Ponies, motoring, relational, hopefulness and generalizations.

word	step	Rule	result
Ponies	1a	ies - i	poni
Motoring	1b	ing - removed	Monitor
relational	2	ational - ate	Relate
hopefulness	3	fulness - ful	hopeful
generalizations	1a	s - removed	generalization

⑤ a) Viterbi algorithm:

$V_1(PRP) = 0.5 \times 0.8 = 0.4$

V_2 = "can" , V_3 = "fish"

MD. $0.4 \times 0.4 \times 0.3 = 0.048$

VB. $0.4 \times 0.5 \times 0.02 = 0.004$

$(0.048 \times 08/0.04 \times 0.1,$

$0.0004 \times 0.4)$

$= 0.000384$

NN $\Rightarrow 0.4 \times 0.1 \times 0.01 = 0.004$

$(0.048 \times 0.2, 0.004 \times 0.5, 0.004 \times 0.3) \times 0.02 = 0.000192.$

NB = The Best final state.

(Back-tracing) they /PRP can/MD fish/VB are with probability of 3.84×10^{-4}.

b) Rule-Based:

Rule-Based has two rules;

Rule 1: If the word is "can" with the tags (MD, VB, NN) and the previous word is personal pronoun and it eliminates NN and VB, Can = MD.

Rule 2: If the previous tag was MD, eliminate NN. so then fish = VB. constant rules automatically eliminates readings from the set until one last tag is left.

④ a)

unigrams		bigrams
<s>	3	(<s>, I) = 2
I	3	I like = 2
like	2	I drink = 1
tea	2	like tea = 1
coffee	1	like coffee = 1
drink	1	drink tea = 1
</s>	3	tea </s> = 2
		coffee </s> = 1

$$P(\text{tea} \mid \text{like}) = \frac{1}{2} = 0.5$$

$$P(\text{drink} \mid \text{like}) = \frac{0}{2} = 0$$

b)

$$P(w \mid v) = \frac{c(v, w) + 1}{c(v) + V}$$

$$P(\text{tea} \mid \text{like}) = \frac{1+1}{2+6} = 0.25$$

$$p(\text{drink} \mid \text{like}) = \frac{0+1}{2+6} = 0.125$$

Reconstituted count:

$$c^*(\text{tea} \mid \text{like}) = P \times c(\text{like})$$

$$c^* = 0.25 \times 2$$

$$c^* = 0.5$$